# KYLE DAWSON, PHD

## Senior Data Scientist | Machine Learning, Uncertainty Modeling, Climate & Geospatial Analytics

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## EXPERIENCE

### Senior Data Scientist — Climate Tech

**SeekOps Inc.**

Mar 2022 – Present Austin, TX

- designed, deployed, and maintained customer-facing analytics platforms (Plotly Dash on Azure and Android) delivering real-time greenhouse gas quantification, QA/QC, and operational decision support securing contracts between $50-150k.
- built and productionized machine learning models to forecast survey quality, detect anomalies, and constrain uncertainty in methane and $CO_2$ emission estimates
- implemented Bayesian uncertainty propagation and boundary-layer physical models to generate decision-grade emissions metrics used in regulatory and enterprise reporting
- developed end-to-end data pipelines integrating sensor data, cloud storage, and model outputs to support scalable analytics and reporting
- authored technical documentation, uncertainty assessments, and customer-facing deliverables supporting multi-year enterprise contracts across 10+ countries

### Data Scientist — Ag Tech

**Bayer CropScience**

Feb 2020 – Mar 2022 St. Louis, MO

- applied machine learning and geospatial analytics to UAV-derived imagery for large-scale digital phenotyping and crop performance assessment
- built end-to-end analytics workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets
- developed interactive Plotly Dash applications delivering insights and decision support to scientific and operational stakeholders
- trained and evaluated deep learning object detection models (Detectron2) for tree species classification from high-resolution aerial imagery
- designed APIs and automated data ingestion pipelines integrating multi-source weather data and mission telemetry
- communicated analytical results and recommendations to cross-functional teams to guide UAS mission planning and optimization

### Research Scientist / Postdoctoral Researcher

**NASA Langley Research Center**

Oct 2017 – Feb 2020 Hampton, VA

- developed algorithms for analyzing vertically resolved aerosol optical properties from High-Spectral Resolution LiDAR (HSRL) data
- built physics-informed retrieval models using optimal estimation and aerosol thermodynamics to estimate particle size distributions and hygroscopicity
- integrated remote sensing observations with trajectory modeling and thermodynamic models to quantify uncertainty in climate-relevant aerosol processes
- performed uncertainty-aware inverse modeling using gradient-based optimization techniques
- produced peer-reviewed publications and technical reports and presented results to scientific and agency stakeholders

## EDUCATION

### PhD, Atmospheric Science

**North Carolina State University**

Graduated 2017 Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing

## EXECUTIVE SUMMARY

Senior Data Scientist with 8+ years of experience building and deploying machine learning, uncertainty quantification, and data analytics systems across climate tech, geospatial analytics, and applied remote sensing. Proven track record of delivering production-grade models, decision-support tools, and scalable data pipelines used by enterprise customers, scientists, and executive stakeholders. Expert in translating complex physical and statistical models into actionable, customer-facing products deployed in cloud environments.

## CORE COMPETENCIES

Stakeholder Communication · Executive Briefings · Technical Leadership · Cross-Functional Collaboration · Decision Support Systems · Product-Oriented Data Science · Requirements Translation · Technical Documentation · Data Storytelling · Model Validation & QA/QC · Uncertainty-Aware Decision Making

## DOMAIN EXPERTISE

Machine Learning · Statistical Modeling · Bayesian Inference · Uncertainty Quantification · Regression Analysis · Geospatial Analytics · Remote Sensing · LiDAR & Satellite Imagery · Time Series Analysis · Anomaly Detection · Spatial Statistics · Environmental & Climate Data

## TOOLS & TECHNOLOGIES

Python · SQL · Git · JupyterLab · Pandas · NumPy · SciPy · Scikit-Learn · Matplotlib · Seaborn · Plotly · AWS S3 · AWS Redshift · Domino Data Lab · GDAL · QGIS · MATLAB · Spotfire · Azure · Docker · APIs

## AWARDS

### NASA Postdoctoral Program Fellow

**NASA Langley Research Center**

Oct 2018 Hampton, VA

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### Langley Aerospace Research Scholar

**NASA Langley Research Center**

Jun 2013 Hampton, VA

## PUBLICATIONS

Google Scholar Profile